FLOODGAUGE · CALCULATION

FloodGauge sizing calculations

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DOCUMENT CONTROL

Document ID	FLG-CAL-001	Version	0.7
Status	Draft	Date	2026-10-31
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/04-calcs/01-sizing.md @ dbb9a87		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-31	Amish Chadha	Kit 1.9.1 upgrade: sizing.py rerun on the kit 1.9 model (171 parts with every fastener); masses and cable runs now measured on it; requirements renamed to PRS numbering (R numbers kept as Was); PR-018 and PR-019 added (section N); STEM check cross-references
0.6	2026-10-02	Amish Chadha	Approved follow-ups: sizing.py rerun with the 120 min pilot R12 target; tamper-resistant fixings priced (cost USD 205.50, USD 45.50 over the target); drain cable slack loop modelled and checked; FieldNode inputs corrected to USD 139.00 and 2.45 kg
0.5	2026-10-02	Amish Chadha	R12 status from the 2026-10-02 decision (120 min for the pilot surface route): met on paper; counts updated. Figures not rerun; sizing.py and results.csv still state the 90 min target
0.4	2026-10-01	Amish Chadha	Constructable design (FLG-DDR-003); arm, bracket and brace loads, mass from the model, cable runs along their

			real routes, cost against the value-engineering target
0.3	2026-09-26	Amish Chadha	Budget approved by Amish ($160); R16 from not met to met on paper
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (range and geometry, depth and level error budgets, tube hydraulics, energy, airtime, alert latency, false readings, arm and clamps, mass, cables, submersion, installation, cost)

FloodGauge sizing calculations

This v0.4 checks the constructable design of FLG-DDR-003 (pole bracket on V-blocks and band clamps, pinned knee brace, head plate, FieldNode at 3.4 m, three hat-section covers, stand-off pipe clamps); sections H, I, J, L and M change, the rest stand. Version 0.2 applied the decisions Amish accepted on 2026-09-25 (FLG-DDR-002): the lens moves to 4.6 m so that it clears a 4.3 m vehicle at the curb, R1 allows 2.5 to 5.0 m set per site, R3 is restated, the pilot cable route runs on the surface under a bolted steel cover, and an M8 through-bolt stops the arm turning. The FieldNode moves up to 3.0 m so that the street head's I2C cable stays short. On 2026-09-26 Amish approved a `budget_usd` of $160 to cover the priced BOM (FLG-DDR-002). On paper, FloodGauge now meets thirteen of its seventeen requirements (ten by calculation, three by design), has three at risk and cannot verify one at TRL 3. R3 is now met on paper. R12 was not met against 90 min: the surface route removes the civil work, but fitting its cover brings the total to 120 min; on 2026-10-02 Amish relaxed R12 to 120 min for the pilot surface route (FLG-DEC-001, item 5), so it is met on paper. The parts added to make the design buildable take the FloodGauge-specific parts to $205.50, $45.50 over the value-engineering target of $160 (a hypothetical control target, not a limit), and the mass on the pole to 4.99 kg, 0.01 kg under R15's 5.0 kg. The three at risk are airtime at slow spreading factors (R7), immersion of the drain head (R9) and the operating environment inherited from FieldNode (R14). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [F2], is the line of that script's output that carries it.

> **Safety:** These are first-principles estimates for a paper proof of concept. They do not show that the gauge will warn anyone in time, and they are no substitute for tests of the heads, the enclosure or the alert chain. FloodGauge supplements official warnings. See FLG-PRC-001, Safety.

Scope and method

The note checks every requirement in FLG-PRS-001 v0.9 (requirements R1 to R17 are PR-001 to PR-017 there) against the design in FLG-PRC-001 v0.6 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS` and derived dimensions, so the head height, arm, tube, cable runs and marker used here are the ones in the STEP files and in drawing FLG-DWG-001 Rev P6. Masses of the made parts and the cable runs are measured on the model. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`, and writes the status table to `docs/04-calcs/results.csv`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is a street with a 150 mm curb and a catch basin 1.3 m deep beside a 60 mm street pole 4.9 m tall above the sidewalk, with tall vehicles in the curb lane, air from -20 to 50 °C, a private LoRaWAN gateway (TwinKit) on EU868, and the FieldNode core as costed and sized in its own repo (FND-CAL-001).

Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Radar head	Distance accuracy ±2 mm; 60 mA at 3.3 V for 0.3 s per reading; half-power beam 25° with a PTFE lens; range up to 20 m	Range from Acconeer; accuracy, current and beam are assumptions, since no data sheet could be checked in this session
Drain head	DFRobot A02YYUW: 3 to 450 cm, ±1 cm, 60° beam, 3.3 to 5 V, 8 mA or less, 100 ms response, -15 to 60 °C, IP67, UART	DFRobot product page, checked 2026-09-25; no built-in temperature compensation is stated, so none is assumed
Air temperature	Speed of sound 331.3 + 0.606 T m/s; calibration at 20 °C; ±3 K between air sensor and air column over the street; ±2 K for an NTC in the drain head	Standard approximation; sensor offsets assumed
Datum	Head height surveyed to ±5 mm (R13 limit); drain head face measured from the grate seat to ±3 mm; steel pole ±35 K from the survey day	R13; tape and level
Surface	±3 mm residual ripple and rain splash after averaging 5 readings	Assumed
FieldNode	15.36 Wh usable; core 4.0 mWh/day at 15 min reports; 16.0 mA s at 3.2 V per SF9 report; rails 90 %; allowance 100 mW design value, 115 mW ceiling; 2.45 kg; $139.00; autonomy at -20 °C is 0.70 of nominal	FND-CAL-001
Controller	5 mA at 3.3 V for 0.35 s per sampling cycle	Typical STM32WL-class figure; assumed
Radio	20-byte payload plus 13 bytes overhead, 125 kHz, CR 4/5, 8-symbol preamble; EU868 1 % duty cycle; The Things Network fair use 30 s/day	FieldNode and TwinKit notes
Network	Gateway, network server and alert service 1 to 4.7 s (TwinKit worst 4.7 s); 2 to 10 s on a city or public server; 1 % uplink loss	TWK-CAL-001; assumed for public servers
Wind and loads	35 m/s gust (q = 750 Pa); drag 2.0 on square tubes and 1.2 on the head; 6063-T6 class aluminium, E = 69 GPa, yield 214 MPa; band clamp preload 1,000 N, friction 0.3; misuse case of a 50 kg person hanging on the head	As FND-CAL-001; handbook values
Vehicles	Tall vehicles 4.0 to 4.3 m at the curb, 0.3 m clearance margin	Typical legal height limits; to confirm with the road authority
Through-bolt	M8 stainless A4-80 (A4-70 before v0.7), stress area 36.6 mm², proof stress 600 MPa in bending as a cantilever, shear 0.6 f_ub; pole wall 2.5 mm, f_u 360 MPa, bearing 2.5 f_u d t	Handbook values; pole wall assumed
Cables	100 pF/m and 0.07 kg/m for outdoor 5-core cable	Assumed

A. Geometry and range (R1, R3)

- **Street range.** With the lens 4.6 m above the road, the radar sees 4.60 m to a dry road and 4.00 m at 600 mm depth, well inside its 20 m range [A1]. Over the whole R1 mounting band of 2.5 to 5.0 m the range runs from 1.9 to 5.0 m; at 5.0 m the assumed beam lights a circle of 1,108 mm radius, so the dry-road background matters more on tall sites [A1b]. R1 is met.

- **Curb echo.** An assumed 25° beam lights a circle of 1,020 mm radius on the road, which includes the curb 250 mm away. If the firmware picked the first echo, the curb top at 4,457 mm would read as 143 mm of water on a dry day [A2]. A beam narrow enough to miss the curb would need to be under 6.2° wide, which a small lens does not give [A3]. The street head therefore needs a recorded dry-road background at installation and tracking of the nadir peak; this is part of R10.
- **Drain range.** The drain head face is 300 mm below the road. The ultrasonic reads levels from 1,250 mm to 330 mm below the road (30 to 950 mm range, within the module's 30 to 4,500 mm), and the wet probe trips at 310 mm [A4]. The 50 mm level lies 10 mm below the tube mouth, in the beam path; the outlet invert is 100 mm above the floor, so the sump always holds water [A6].
- **R3 as restated.** Between 330 mm and 40 mm below the road (the grate underside), a 290 mm band, the only signal is the wet probe at one point [A5]. The drain head cannot rise higher, because the road slab and grate frame sit above it. In practice a basin in that band is surcharging, the wet probe says so, and the street head reads the water once it reaches the road. Under FLG-DDR-002, R3 now asks for a continuous reading from 50 mm above the basin floor to 330 mm below the road, plus a drain-full signal above that, so R3 is met on paper.
- **Vehicle clearance.** At 2.95 m (v0.1) the head was inside the envelope of a 4.0 to 4.3 m vehicle parked or passing at the curb. Under FLG-DDR-002 the reference lens height is 4.6 m, which clears a 4.3 m vehicle with the 0.3 m margin [A7]. Each site sets its own height within 2.5 to 5.0 m to the road authority's clearance rule. The site needs a pole about 4.9 m tall above the sidewalk; a shorter sign pole only suits a curb lane without tall vehicles.

B. Street depth error (R2)

- **Radar.** The error budget is module ±2.0 mm (assumed), datum ±5.0 mm, pole expansion ±1.98 mm (the arm is now 4.71 m up) and surface ripple ±3.0 mm, which gives ±6.5 mm root sum square and ±12.0 mm if every term is at its worst at once [B1]. R2 (±10 mm) is met on paper by the root sum square, with the module accuracy still unverified.
- **Ultrasonic variant, uncompensated.** Calibrated at 20 °C, the range error is +349 mm at -20 °C and -231 mm at 50 °C, because the speed of sound runs from 319.2 to 361.6 m/s [B2].
- **Ultrasonic variant, compensated.** With an air sensor within ±3 K of the air column, the temperature term is ±24.4 mm and the total ±27.0 mm at the 4.6 m height [B3]. R2 is not met by the ultrasonic variant, which confirms the choice of radar (D3). Sun on the road makes the ±3 K figure itself optimistic.
- **Arm deflection.** The 40 x 40 x 1.6 mm arm (I = 60,503 mm⁴) moves 0.22 mm under a 1 kg bird at the head even without the brace [B4]; negligible.

C. Drain level and stilling tube (R4, R3)

- **Error budget.** With the NTC in the drain head (±2 K) at the full 950 mm range, the temperature term is ±3.4 mm and the total ±11.0 mm [C1]. Without it (±10 K from a 20 °C calibration) the temperature term is ±16.8 mm and the total ±19.7 mm, just inside the target with no margin [C2]. R4 (±20 mm) is met on paper, and the NTC (BOM line 4, $0.50) is kept for margin.
- **Slots.** The 69 mm bore has 3,739 mm² of area. Forty slots of 5 x 50 mm in ten rows give 1,000 mm² per row and 10,000 mm² in all, 4.5 % of the wall [C3].
- **Lag.** With only the lowest row of slots wet, the level inside the tube lags the basin by 0.00 mm at 20 mm/min, 0.20 mm at 10 mm/s and 4.95 mm at an extreme 50 mm/s [C4]. The tube does not slow the reading; its job is to give the wide ultrasonic beam a clean path and to calm the surface.

D. Energy (R8)

- **Per cycle.** A radar reading takes 59.4 mJ and an ultrasonic reading 12.0 mJ; with 90 % rail converters and 5.8 mJ for the controller, a cycle costs 85.1 mJ [D1].
- **Daily draw.** Normal sampling every 60 s is a 1.42 mW sensor load and 0.038 Wh/day with the FieldNode core. Event sampling every 10 s all day, with an uplink every minute, is 8.51 mW for the sensors plus 0.80 mW for the extra uplinks, 0.227 Wh/day in all [D2].
- **Against FieldNode.** Event sampling all day uses 9.3 % of the 100 mW FieldNode design allowance (115 mW ceiling) [D3]. The cell lasts 68 days without sun, or 47 days at -20 °C [D4]. R8 (5 days) is met with a wide margin.
- **Heat.** FieldNode's R3 finding (the cell too hot to charge on clear hot days) matters less here: even the 0.8 Wh a hot clear day stores without a shield exceeds the 0.227 Wh drawn [D5].

E. Airtime (R7)

Table 2. Airtime for a 20-byte reading [E1].

Spreading factor	Airtime per uplink	Normal mode (15 min)	Event mode (1 min) against 1 % (36 s/h)	Shortest interval under 1 %	Event minutes before The Things Network's 30 s/day
SF7	71.9 ms	6.9 s/day	4.3 s/h, met	7 s	344
SF8	133.6 ms	12.8 s/day	8.0 s/h, met	13 s	138
SF9	246.8 ms	23.7 s/day	14.8 s/h, met	25 s	27
SF10	452.6 ms	43.5 s/day	27.2 s/h, met	45 s	0 (normal mode already over)
SF11	987.1 ms	94.8 s/day	59.2 s/h, **not met**	99 s	0
SF12	1,810.4 ms	173.8 s/day	108.6 s/h, **not met**	181 s	0

- **R7 is at risk.** On a private gateway the 1 % duty cycle holds in event mode from SF7 to SF10, but not at SF11 or SF12, where the firmware must stretch event reporting to 99 s and 181 s. On The Things Network, an SF9 gauge in normal mode uses 23.7 of its 30 s a day and has room for only about 27 min of event reporting; at SF10 and slower even normal mode breaks the fair use limit (FieldNode R9). This supports D7: event reporting goes through a private or city gateway.

F. Alert latency (R6)

The chain is: wait for the next sample, two confirming samples 10 s apart, about 1 s to wake and queue, the uplink, a possible duty-cycle wait of up to 24.4 s after a previous uplink, a repeat after a lost packet (1 %), and the gateway, network server and alert service.

Table 3. Alert latency from a band crossing to the message leaving the alert service, 200,000 simulated events [F1].

Network	Case	Mean	95th percentile	Largest simulated
Private gateway	Slow rise from normal mode (60 s sampling)	55 s	82 s	125 s
Private gateway	Already in event mode (10 s sampling)	34 s	52 s	84 s
City or public server	Slow rise from normal mode	58 s	85 s	134 s
City or public server	Already in event mode	38 s	55 s	89 s

- **R6 is met on paper at SF9 and SF10.** The 95th percentile is 82 s on a private gateway and 85 s on a city server, against 120 s. The deterministic worst case without a lost packet is 110 s (60 + 20 + 1.2 + 24.4 + 4.7 s) [F2]; only a lost packet pushes an event past 120 s. The independent review pointed out that this holds only to SF10: in event mode on a private gateway the 95th percentile is 71 s at SF10, 123 s at SF11 and 201 s at SF12 [F4], so slow spreading factors need a shorter report interval or a nearer gateway (as for R7).
- **Depth at alert.** Water rising at 20 mm/min adds 37 mm between the crossing and the alert; at 100 mm/min, when the gauge is already sampling every 10 s, it adds 101 mm in 60 s [F3]. The first alert band (100 mm, set on 2026-10-02) therefore warns at roughly 140 to 200 mm in a fast flood; the 150 mm band warns at roughly 190 to 250 mm.
- The TRL 2 figure of about 70 s typical and 110 s worst stands; the new 95th percentile is the figure R6 asks for.

G. False readings (R10)

- **Tall objects.** A car roof (1.45 m), a van roof (2.00 m) and a person (1.75 m) under the head read as depths well above 600 mm and are rejected as out of range. A 0.5 m snow bank reads as 500 mm and needs the drain cross-check [G1].
- **Rules defined.** Reject depths above 600 mm; reject a step above 50 mm between 10 s samples unless the drain head is full or rising; require three consecutive readings above a band; subtract the dry-road background recorded at installation (A2); and flag a gauge that has been silent for 30 min [G2].
- **Status.** The rules cover the obvious cases, but the target (one false alert per gauge per year or fewer) can only be checked from a field log. R10 is not verifiable at TRL 3.

H. Arm, brace, pole bracket and clamps (R15)

- **Load path (FLG-DDR-003).** The arm is held between two angle cleats on a 3 mm bracket plate that sits on the pole on two V-blocks and two band clamps 400 mm apart; a 25 x 25 x 1.6 mm brace is pinned between a clip under the arm and a clip on the plate, 358 mm between pins at 45°. The arm carries pull into the cleats and the brace pushes on the plate above the lower band.
- **Wind twist.** Wind along the street at 35 m/s puts 43.0 N on the arm, 6.2 N on the head and 13.4 N on the brace, twisting the arm about the pole with 24.8 N·m [H1]. Two band clamps at 1,000 N preload hold 36.0 N·m by friction, a factor of only 1.45 [H2]. A twisted arm moves the head along the street, not up or down, so it does not bias the depth.
- **Anti-rotation bolt.** The M8 through-bolt runs across the street through the bracket plate, a 20.2 mm spacer and both pole walls (a band clamp cannot take a bolt). Version 0.6 treated it as a couple across the full 60 mm pole (922 N·m, a factor of 37); the independent review of the STEM check showed that is the wrong load path. The wind twist presses the plate sideways on the bolt, 52 mm from the pole axis, so the bolt bends as a 21.7 mm cantilever out of the near pole wall. In class A4-80 stainless (proof stress 600 MPa) it yields at 862 N, which

resists 45 N·m, a factor of 1.8 against the 24.8 N·m wind twist; shear (17.6 kN) and pole wall bearing (18.0 kN with an assumed 2.5 mm wall) do not govern [H2b]. The class was raised from A4-70 (33 N·m, factor 1.3) under FLG-DDR-004 F2. On a 40 mm pole friction alone gives 24.0 N·m, a factor of only 0.97, so there the through-bolt carries the twist [H2c]. Drilling the pole needs the asset owner's permission.

- **Misuse.** A 50 kg person hanging on the head gives 181 N·m at the top brace pin and 60 MPa in the 1.6 mm arm wall against 214 MPa (factor 3.6); the brace carries 1,504 N against 11,520 N, set by bearing in its 1.6 mm walls at the pins; Johnson buckling gives 28,533 N (L/r is 37, so the Euler load of 73,043 N used in v0.6 does not apply) [H3]. The arm pulls on the cleats with 1,063 N, carried by the upper band against 2,000 N (factor 1.9), and 1,200 N of slip resistance holds 520 N down [H4].
- **Fixings.** M6 A4-70 bolts carry 8.4 kN per shear plane: the arm cleat joint (two bolts in double shear) holds 34 kN and the 1.6 mm arm walls 23.0 kN in bearing against the 1.06 kN pull; each brace pin in double shear holds 16.9 kN against 1.50 kN [H5].

I. Mass on the pole (R15)

- **Items.** FieldNode 2.45 kg (its constructable design, FND-BLD-001); from the model: bracket plate 0.27 kg, V-blocks 0.14 kg, cleats 0.05 kg, brace clips 0.07 kg, arm 0.46 kg, brace 0.15 kg, head plate 0.08 kg, band clamps 0.04 kg, through-bolt and spacer 0.06 kg, M6 fixings, sleeves, pins and head screws 0.27 kg, marker plate 0.21 kg and its band clamps 0.06 kg; radar head 0.25 kg (assumed); street cable 0.21 kg (3 m); drain cable on the pole 0.22 kg [I1].
- **Total.** 4.994 kg against 5.0 kg, a margin of 0.006 kg or 0.1 % (v0.3: 4.74 kg) [I2]. R15 is met on paper, with a margin within the accuracy of the estimate; the arm and brace walls went from 2.0 to 1.6 mm and the bracket plate has two windows to hold it. The riser guard and the three covers (2.00 kg) stand on the sidewalk and are not carried by the pole. Weighing the prototype is a TRL 4 check (FLG-DEC-001, open decision 3).

J. Cables and ports (R11 context)

In v0.7 the model draws each cable ending in its plug moulding 45 mm above the node base, which shortens the street run from 2,641 to 2,619 mm and the drain run from 4,335 to 4,310 mm; no design dimension changed (FLG-DDR-004, Part 3).

- **Street cable.** Measured along its real route on the model (from the head's side gland along the arm, down the pole outside the band clamps and up to port A under the FieldNode box), the run is 2,619 mm with the FieldNode centre at 3.4 m; with 300 mm of drip loops (2.92 m) it fits a 3 m cable, and the I2C bus is about 350 pF against 400 pF [J1]. With the node at 3.0 m the same route needed 3.34 m with loops, which is why the node moved up (FLG-DDR-003, P6).
- **Drain cable.** Up through the grate opening, under the three covers, up the riser guard and the pole to port B, the run is 4,310 mm; with 500 mm of loops (4.81 m) it fits the 5 m cable, which carries UART and the analog wet probe [J2].
- **Ports.** The street head uses port A at 3.3 V (I2C) and the drain head port B at 5 V (UART plus the analog pin), consistent with FieldNode's one switched rail per port [J3].

K. Submersion (R9)

- Water 600 mm above the road puts 0.90 m (8.8 kPa) over the drain head face; R9 asks for 2.0 m (19.6 kPa) for 72 h, while the stock module is rated IP67 (1 m for 30 min) [K1]. The electronics can be potted, but the

transducer face cannot be potted over, so its own seal decides. R9 is at risk until a longer immersion rating or a test shows otherwise.

L. Installation (R12)

- **Surface work.** Traffic protection 10 min, the arm and bracket (assembled on the ground) lifted and clamped from a mobile platform 15 min, anti-rotation bolt 5 min, FieldNode and street head 15 min, marker 5 min, survey and dry background 10 min, grate lift 5 min, two wall anchors drilled through the grate opening 20 min, tube and head 10 min, the three covers, riser guard and marker clamps 25 min: 120 min in all [L1].
- **Cable route (FLG-DDR-002).** For pilots the drain cable comes up through the grate opening nearest the curb (FLG-DDR-003, P7) and runs under 0.76 m of bolted steel hat-section cover across the gutter strip, up the curb face and across the sidewalk to a riser guard at the pole. The anchors go into the curb and sidewalk only, none into the road [L2]. The conduit (coring the basin wall and a 0.50 m trench) is kept for permanent sites installed with road works.
- **R12 is met on paper for the pilot route:** 120 min with no civil work, against the 120 min that Amish set for the pilot surface route on 2026-10-02 (FLG-DEC-001, item 5); 90 min stays the target at permanent sites on the conduit route.

M. Cost (R16)

- The BOM has 10 lines, all priced. Value-engineering target: USD 160. Estimated cost of the constructable design: USD 205.50 for the FloodGauge-specific parts (lines 2 to 10), USD 45.50 over the target; with the USD 139.00 FieldNode core (its constructable design), a complete gauge costs USD 344.50 [M1]. The pole bracket, cleats, clips, head plate and their fixings (line 3, USD 20.00 to 35.00), the three covers and riser guard (line 7, USD 14.00 to 26.00), the stand-off pipe clamps (line 5, USD 14.00 to 18.00), the marker band clamps (line 8, USD 6.00 to 10.00) and the drain head socket (line 4, USD 22.50 to 24.50) account for the rise from USD 153.50. The main cost drivers are the radar head, the arm and bracket, the covers and the drain head [M2]. R16 is reported against the value-engineering target; savings worth trying are listed in FLG-DEC-001. Radar module, housing and cover prices are indicative.

N. Tamper resistance and cover profile (PR-018, PR-019)

- **Security fixings.** Eight masonry anchors with snake-eye security nuts and two marker band clamp screws with pin-Torx heads: 10 of the 10 fixings within reach of the sidewalk need a security tool (BOM items 7, 8 and 10) [N1]. PR-018 is met by design.
- **Cover profile.** The surface covers are a 2 mm hat section 16 mm high, the target for PR-019, and the kit 1.9 model breaks every exposed edge (`python .kit/cad_check.py`) [N2]. Met by design.

Results

Table 4. Requirement status (also written to `docs/04-calcs/results.csv` *).*

ID	Requirement	Value	Target	Status
R12	Installation	120 min on the surface route; no civil work	120 min on the pilot surface route (90 min at permanent sites); no basin entry	Met on paper
R7	Airtime	23.7 s/day at SF9; event 1 % met SF7 to SF10, not SF11 or SF12	1 % always; TTN 30 s/day in normal mode	At risk
R9	Drain head submersion	Potted head; transducer face seal IP67 only	IP68, 2 m, 72 h	At risk
R14	Operating environment	FieldNode R2 and R3 heat findings (inherited); A02YYUW rated -15 to 60 °C	-20 to 50 °C	At risk
R10	False readings	Rules defined; curb echo needs a recorded background	1 false alert per year or fewer	Not verifiable at TRL 3
R1	Street depth range	Lens 4.60 m clears a 4.3 m vehicle; range 1.9 to 5.0 m over the band; module to 20 m	0 to 600 mm; head 2.5 to 5.0 m per site	Met on paper
R2	Street depth accuracy	Radar ±6.5 mm (sum ±12.0 mm), module accuracy assumed; ultrasonic variant ±27.0 mm	±10 mm	Met on paper
R3	Drain level range	Continuous from 1,250 to 330 mm below the road; wet probe at 310 mm	50 mm above floor to 330 mm below road, drain-full above	Met on paper
R4	Drain level accuracy	±11.0 mm with NTC; ±19.7 mm without	±20 mm	Met on paper
R6	Alert latency	82 s at the 95th percentile; 110 s worst without packet loss at SF9; event-mode 95th percentile 123 s at SF11 and 201 s at SF12	120 s, 95 % of events	Met on paper (private gateway, SF9 and SF10)
R8	Energy autonomy	68 days nominal, 47 days at -20 °C	5 days at event sampling	Met on paper
R13	Level datum	Tape and radar dry-background survey; yearly marker check	±5 mm	Met on paper
R15	Mounting	4.994 kg (margin 0.006 kg, inside the accuracy of the estimate); twist factor 1.8 on the through-bolt, 1.45 on friction (60 mm pole)	40 to 60 mm poles; 5 kg	At risk
R16	Cost	$205.50 FloodGauge-specific; $344.50 with FieldNode	$160 FloodGauge-specific (value-engineering target)	Over the value-engineering target by $45.50
R5	Sampling rate	60 s and 10 s schedule in the sampling logic	60 s; 10 s in events	Met by design
R11	Privacy	Levels, status and battery only	No camera or microphone	Met by design

ID	Requirement	Value	Target	Status
R17	Open data	JSON or CSV through the gateway	Open format	Met by design
PR-018	Tamper resistance	10 of 10 fixings within reach need a security tool	All fixings within reach	Met by design
PR-019	Cover profile and edges	Hat section 16 mm high, edges broken	16 mm or less	Met by design

Counts: 0 not met, 4 at risk (R7, R9, R14, R15), 1 not verifiable at TRL 3, 8 met on paper, 5 met by design (R5, R11, R17, PR-018, PR-019); R16 is over the value-engineering target by $45.50. Before the 2026-10-02 decision on R12, R12 was not met.

Changes in v0.7 (kit 1.9.1 upgrade, 2026-10-31)

- `sizing.py` rerun on the kit 1.9 model (`components("A")`: 171 parts with every bolt, nut, washer and plug). Masses and cable runs are measured on it: mass on the pole 4.994 kg (margin 0.006 kg), street cable run 2,619 mm, drain cable run 4,310 mm. All other figures are unchanged.
- Requirements are cited by their R numbers, which are the "Was" numbers of PR-001 to PR-017 in FLG-PRS-001; PR-018 and PR-019 are new (section N).
- The independent review in FLG-STM-001 re-derived the headline figures. It refuted the through-bolt couple (922 N·m, factor 37): the bolt bends, so the factor is 1.8 with class A4-80 (section H, FLG-DDR-004 F2). It also showed that the brace load used the Euler formula where it does not apply (Johnson and pin bearing now used), that R6 holds only to SF10 (F4), and that the 6 g mass margin is inside the estimate's accuracy (R15 is now at risk). Version 0.6 text on the section M line prices and the 150 mm band was out of date and is corrected.

Changes in v0.6 (approved follow-ups, 2026-10-02)

- `sizing.py` rerun with R12 at 120 min for the pilot surface route (90 min stays the target at permanent sites); `results.csv` now agrees with the text.
- Tamper-resistant fixings priced: line 7 USD 26.00 to 32.00, line 8 USD 10.00 to 13.00, line 10 USD 6.00 to 12.00. FloodGauge-specific cost USD 190.50 to 205.50, now USD 45.50 over the target.
- Drain cable 0.5 m slack loop added to the model, tied to the upper pipe clamp, with clearance checks against the tube, its slots, the drain head, the basin wall and the clamps. The cable runs and mass on the pole are unchanged.
- FieldNode inputs table corrected to USD 139.00 and 2.45 kg.

Changes in v0.4 (FLG-DDR-003, design for construction)

- Arm and bracket: the arm (now 715 mm, 40 x 40 x 1.6 mm) sits in cleats on a pole bracket plate on V-blocks; the brace (25 x 25 x 1.6 mm) is pinned in two clips; the through-bolt runs across the street. Misuse case: arm stress 36 to 60 MPa (factor 5.9 to 3.6), band pull 858 to 1,063 N (factor 2.3 to 1.9). Twist factor 37 unchanged.
- Mass on the pole 4.74 to 4.99 kg, now measured on the model; R15 still met on paper with a 0.01 kg margin.
- FieldNode centre 3.0 to 3.4 m; street cable run 2.64 m along its real route (2.94 m with loops in 3 m); drain cable 4.34 m (4.83 m in 5 m).

- Cost USD 153.50 to 205.50 FloodGauge-specific; R16 now reported against the value-engineering target (USD 45.50 over). FieldNode core USD 126.00 to 139.00 and 2.41 to 2.45 kg (its constructable design).

Changes in v0.3 (budget approved)

- `budget_usd` $150 to $160, approved by Amish on 2026-09-26 (FLG-DDR-002). R16 not met to met on paper, $6.50 margin.

Changes in v0.2 (FLG-DDR-002)

- Lens height 2.95 m to 4.6 m; R1 band 2.5 to 3.5 m to 2.5 to 5.0 m, set per site. R1 stays met; the vehicle clearance problem is closed.
- FieldNode center 2.35 m to 3.0 m; street cable 2 m to 3 m (I2C about 250 to 350 pF).
- R3 restated; not met to met on paper.
- Cable route: conduit (civil work) to a surface cover for pilots. R12 installation 90 min plus civil work to 120 min with none; still not met.
- Anti-rotation M8 through-bolt: twist factor 1.45 to 37.
- Mass on the pole 4.62 kg to 4.74 kg.
- FloodGauge-specific cost $145.50 to $153.50; complete gauge $271.50 to $279.50. R16 met to not met.
- Street depth error, radar ±6.3 mm to ±6.5 mm (pole expansion over the taller height).

Changes to the TRL 2 figures (v0.1)

- Parts cost: FloodGauge-specific $145.50 (was about $145) after adding an NTC to the drain head; complete gauge $271.50 (was about $271).
- Mass on the pole: 4.62 kg (was about 3.1 kg), because FieldNode is now 2.41 kg.
- FieldNode center lowered from 2.5 m to 2.35 m above the road in the model, so that its panel clears the knee brace (drawing FLG-DWG-001, Detail A).
- Alert latency: 82 s at the 95th percentile (TRL 2 gave 70 s typical and 110 s worst, which stand).
- Airtime: event reporting on The Things Network lasts about 27 min at SF9 (TRL 2 said about 24 min).
- R3 was "met by design" at TRL 2 and is not met; R12 was "at risk" and is not met.